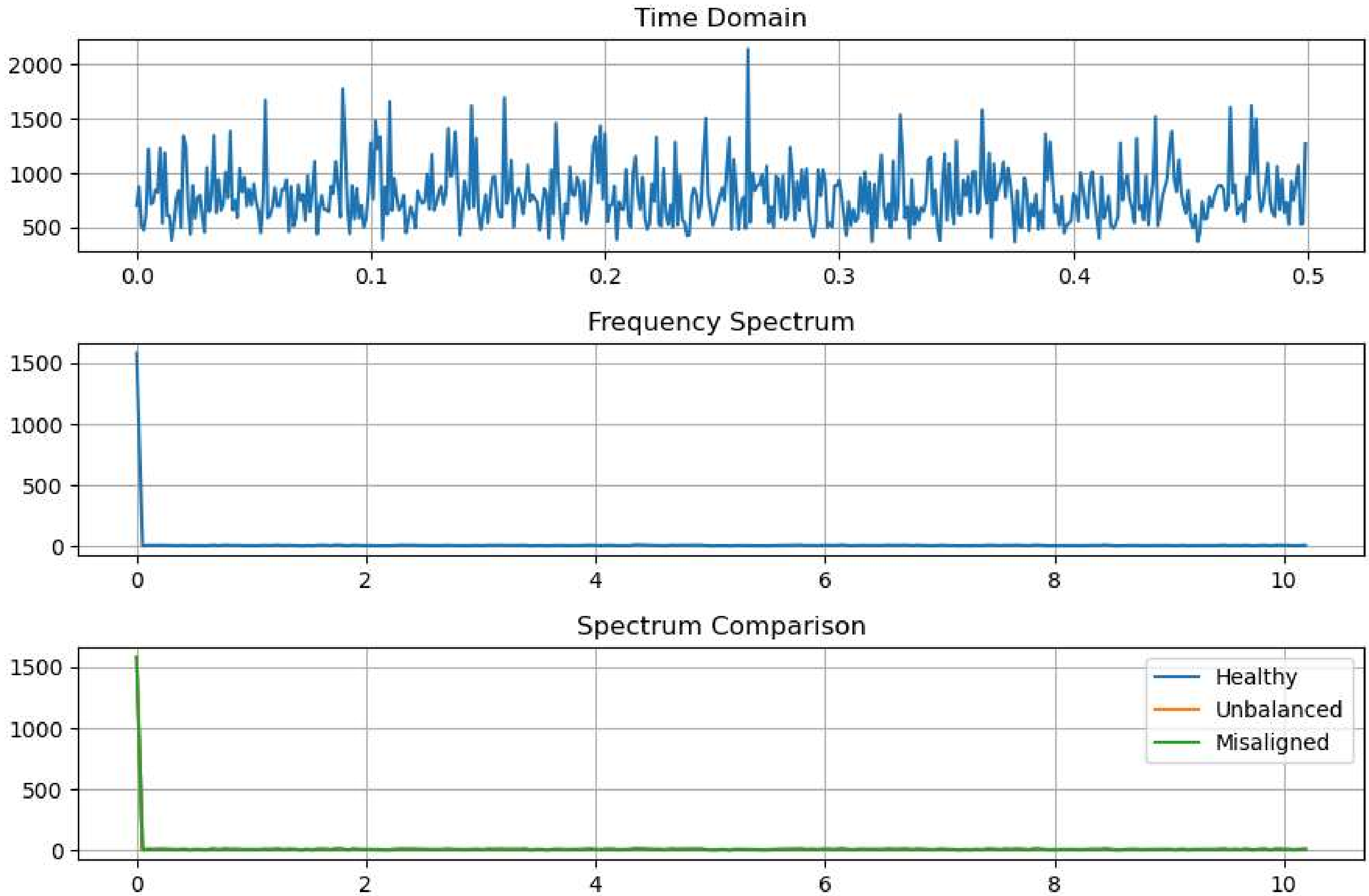
In [1]:

```
import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
from scipy.fft import fft, fftfreq
import os

path = r"C:\Users\shreya hajare\Downloads\Engine Data.zip"

df = pd.read_csv(path)

print(df.head())

x = df.select_dtypes(include='number').iloc[:, 0].values

fs = 1000
t = np.arange(len(x)) / fs
N = len(x)

f = fftfreq(N, 1/fs)[:N//2]
F = lambda s: 2/N * np.abs(fft(s)[:N//2])

u = x + 0.5 * np.sin(2 * np.pi * 25 * t)
m = x + 0.7 * np.sin(2 * np.pi * 100 * t)

plt.figure(figsize=(9, 6))

plt.subplot(3, 1, 1)
plt.plot(t[:500], x[:500])
plt.title("Time Domain")
plt.grid()

plt.subplot(3, 1, 2)
plt.plot(f[:200], F(x)[:200])
plt.title("Frequency Spectrum")
plt.grid()

plt.subplot(3, 1, 3)
plt.plot(f[:200], F(x)[:200], label="Healthy")
plt.plot(f[:200], F(u)[:200], label="Unbalanced")
plt.plot(f[:200], F(m)[:200], label="Misaligned")
plt.legend()
plt.title("Spectrum Comparison")
plt.grid()

plt.tight_layout()
plt.show()

rms = np.sqrt(np.mean(x**2))

cond = "HEALTHY" if rms < 0.5 else "UNBALANCED" if rms < 1 else "MISALIGNED/FAULTY"

print("\nVIBRATION ANALYSIS RESULT")
print("RMS Value:", round(rms, 4))
print("Engine Condition:", cond)
```

```
   Engine rpm  Lub oil pressure  Fuel pressure  Coolant pressure  \
0         700          2.493592      11.790927          3.178981
1         876          2.941606      16.193866          2.464504
2         520          2.961746       6.553147          1.064347
3         473          3.707835      19.510172          3.727455
4         619          5.672919      15.738871          2.052251

   lub oil temp  Coolant temp  Engine Condition
0     84.144163     81.632187                 1
1     77.640934     82.445724                 0
2     77.752266     79.645777                 1
3     74.129907     71.774629                 1
4     78.396989     87.000225                 0
```




```
VIBRATION ANALYSIS RESULT
RMS Value: 835.2674
Engine Condition: MISALIGNED/FAULTY
```

In [ ]: